

**Fifth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

ELECTROMAGNETIC WAVES

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
 - (2) All questions carry marks as indicated.
 - (3) Illustrate your answers with neat sketches.
 - (4) Assume suitable data wherever necessary.

1.
 - (a) Derive the expression for the input impedance of transmission lines in terms of load impedance and characteristic impedance. 5(CO1)
 - (b) An impedance of $75 + j30 \Omega$ is connected to 100Ω line. Find the complex reflection coefficient at the load, the complex reflection coefficient and impedance at a distance of 0.25λ from the load. What is the VSWR measured on the line ? 5(CO1)

2.
 - (a) Derive expressions for attenuation constant and phase constant for uniform plane waves. 5(CO2)
 - (b) A uniform plane wave at frequency of 300 MHz travels in vacuum along +y direction. The electric field of the wave at some instant is given as $\mathbf{E} = 3 \mathbf{a}_x + 5 \mathbf{a}_z$. Find the phase constant of the wave and also the vector magnetic field. 5(CO2)

3.
 - (a) Explain the behavior of plain waves at dielectric interface. 5(CO4)
 - (b) A light beam is incident from air to a medium with a dielectric constant 4 and relative permeability 100. If the angle of incidence is 60° , find the angle of reflection and angle of refraction. 5(CO4)

4. (a) Formulate the wave equations for TE waves in parallel plane waveguide. Show that the waveguide behaves like a high pass filter. 6(CO5)
- (b) An electromagnetic wave of frequency 300 Hz is guided by a pair of parallel copper planes in z direction. The separation between the planes is 10 cm. Calculate the guide wavelength. 4(CO5)
5. (a) Explain the impossibility of TEM waves in rectangular wave guides. 4(CO5)
- (b) A rectangular waveguide measuring 3 cm × 4.5 cm internally has 9 GHz signal propagating in it. Calculate cut off wavelength, guide wavelength, phase velocity, group velocity and characteristic wave impedance for $TE_{1,0}$ mode. 6(CO5)
6. (a) Explain the following terms related to antenna :
- (i) Front to back ratio.
 - (ii) Antenna Beam width.
 - (iii) Directivity.
 - (iv) Antenna Gain.
 - (v) Half power beamwidth. 5(CO3)
- (b) Transmitting and receiving antennas operating at 2 GHz with gains of 25 dB and 20 dB respectively are separated by a distance of 2 km. Assume that both antennas are identical in polarization. Estimate the received power if the transmitted power is 150 W. 5(CO3)

